Empirical Evaluation of Vibration-Only Acoustic Emission for Early Bearing Wear: Cloud vs On-Premise

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ABSTRACT

The necessity of complex multimodal imaging for industrial bearing diagnosis remains disputed. In this empirical study, we evaluate high-frequency acoustic emission (100 kHz - 1 MHz) alongside 3-axis vibration across 80 industrial motors. Our findings reveal that acoustic emission provides reliable warning 48 hours prior to catastrophic failure without requiring optical cameras. We show that optical thermal cameras add minimal value in enclosed gearboxes while introducing severe maintenance issues. However, models suffer a 34.2% accuracy drop when transferred across different motor horsepower ratings.

1. Introduction

Several recent works argue for multimodal computer vision in predictive maintenance. We hypothesize that for enclosed machinery such as industrial pumps and planetary gearboxes, acoustic emission captures sub-surface crack propagation far earlier and at lower cost.

2. Testbed & Empirical Setup

We monitored 80 induction motors ranging from 15kW to 75kW under fluctuating mechanical load profiles. Acoustic emission was captured using resonant piezoelectric sensors with cloud batch streaming.

3. Findings & Comparative Analysis

Across all evaluated speed regimes, high-frequency acoustic emission and 3-axis vibration data provided sufficient early fault indicators 48 hours prior to catastrophic failure; optical and thermal camera streams added negligible discriminative accuracy (<0.4% gain) while introducing severe occlusion vulnerabilities in enclosed oil-lubricated gearbox housings.

4. Domain Transfer Limitations

A critical unaddressed challenge is cross-machine domain transfer: models trained on 15kW induction motors exhibited a 34.2% drop in anomaly detection accuracy when tested on 75kW compressor pumps without extensive target domain recalibration. Current supervised models fail to generalize across differing mechanical resonant frequencies and operational loads.

5. Summary

Vibration and acoustic emissions remain the most practical modality for enclosed machinery. Research must focus on unsupervised domain adaptation rather than adding superfluous optical camera hardware.